**R13**

**Code No: 117BX**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech IV Year I Semester Examinations, November/December - 2018**
**COMPUTER GRAPHICS**
**(Common to CSE, IT)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

**PART- A**

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | Define Pixel Addressing. | [2] |
| b) | List out different Input and Output Devices. | [3] |
| c) | Differentiate between window and view port. | [2] |
| d) | Describe Reflection. | [3] |
| e) | Define Parallel Projection. | [2] |
| f) | Differentiate 2D and 3D Transformations. | [3] |
| g) | What are the polygon rendering methods? | [2] |
| h) | Explain the steps involved in phong shading algorithm. | [3] |
| i) | What is meant by scripting system? | [2] |
| j) | What are features of Morphing? | [3] |

**PART-B**

**(50 Marks)**

2. Explain DDA Line generation algorithm with an example. [10]

**OR**

3.a) Differentiate between Raster Scan and Random Scan Display Units.
b) Explain about the beam penetration method. [5+5]

4. What are composite transformations? Describe the rotation and Reflections with suitable illustrations. [10]

**OR**

5. Describe about Cohen Sutherland Line Clipping Algorithm with example. [10]

6. Explain the following
a) Quadric Surfaces b) 3D clipping. [5+5]

**OR**

7. Describe different Design Techniques using Bezier Curves. [10]

8. Explain different Visible-Surface Detection Algorithms. [10]

**OR**

9. Discuss different Illumination Models to calculate Light Intensities. [10]

10. Explain step by step design features to implement Animation. [10]

**OR**

11. Write a program to implement motion of a bouncing ball using a downward gravitational force and ground plane friction force. Initially, the ball is to be projected into space with a given Velocity vector. [10]

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